



do governments that spy on each other collaborate on standards for the IoT ecosystem?

At the IEW panel discussion on IoT security, David lauded the Indian government's efforts in trying to bring about standardisation within the IoT space. In 2017, Telecom Regulatory Authority of India (TRAI) presented Department of Telecommunications (DOT) with its recommendations on Security by Design principles to be followed by IoT device makers.

Another panellist at the IEW talk, Tulika Pandey, director, Computer Emergency Response Team-India (CERT-In), MeitY, spoke of another government initiative—the opening of four centres of excellence for the IoT. The first one came up in Bengaluru and spawns over 22 IoT startups, four of which are fully-operational.

Talking about the importance of standards to support security, she also highlighted CERT-In's Reference Architecture for Smart Cities that IoT firms and independent designers should follow. She went on to invite stakeholders to interact with the government on the upcoming Data Protection Act.

Are standards being developed in silos

The world is now a village, particularly when it comes to tech. The latest smartphone launched in, say, South Korea, is available in the remote corners of India, and operates seamlessly on Indian telecom networks. So IoT security standards being developed need to be global to ensure security at every level. Yet, it seems that there is no single concerted global effort—something that ought to be driven by the UN, IEEE or the EU, or a combination of the three.

There is IoT Security Foundation, a UK-driven body with representatives from British Telecom, Vodafone, Cisco and others. This body has worked closely with the UK government on developing standards and best practices focussing on security. But it is still evolving and, at the moment, inviting participation from senior management in big telecom and component manufacturing firms, particularly from the US and Far East, which are under-

Some IoT Security Standards Developed Across Verticals

Automobile: SAE J 3061

Aerospace: DO 326/DO 356

Industrial IoT: ISA/IEC 62443 (derived from IEC 61508)

represented in this forum.

Meanwhile, National Institute of Standards and Technology (US Department of Commerce) is working separately on IoT standards, with members from the US telecom and manufacturing sector. With all these disparate, disconnected efforts, are we heading from a proliferation of devices to a proliferation of standards?

In his talk, Binoy mentioned that, according to F&S' analysis, for IoT revenues of, say, US\$ 200 billion, the actual value that the IoT would deliver could be pegged at US\$ 7.2 trillion. This would accrue from higher efficiencies, improved productivity, quicker time-to-market, improved logistics and so on. But this computation is again based on assuming a level of seamlessness and assured security levels.

The flip side to this calculation is trillions of dollars in losses, if IoT devices result in data breaches, car accidents (vehicles will soon have over 200 sensors), flight delays, mistimed deliveries and the like.

Device duplication

Starry-eyed projections on how big the IoT market can grow also depends on factors that have not been resolved yet. Globe trotters know that airport shops sell chargers and converters—a necessary evil because devices manufactured for India, most parts of Asia and all of Europe cannot be plugged into US sockets, and US devices can only work in other parts of the world with the help of converters.

So economies of scale, possible when manufacturing for a truly global market, are negated by the need for the same device to be manufactured differently, to meet different countries' electricity specs.

Sivakumar Natarajan, global practice head - digital, manufacturing and

technology business unit, Wipro, highlighted the fact that companies have moved beyond delivering products or devices. Rather, they now offer platforms. Taxi aggregators like Ola and Uber do not own cabs but make money on their platforms. Swiggy does not own restaurants but monetises others' assets via a smart service offered over a platform. Airbnb owns no hotel rooms but offers accommodation across the world from its site.

Hence, with the arrival of platforms, seamless and secure integration of devices, interfaces and networks are even more important. Natarajan referred to this as the ecosystem economy. In such an economy, interoperability depends entirely on widely-accepted standards. And the more globally accepted the standards are, the closer the ecosystem economy will get to realising the true value of the IoT, which F&S has pegged at US\$ 7.2 trillion (on revenues of US\$ 200 billion).

Reasons for optimism

A healthy note of optimism was struck by Deepu Chandran, technology manager, LDRA Tech, another IEW speaker. He reminded the audience about how the aerospace industry evolved. In this high-stakes, mission-critical domain, standards evolved in the early 1990s. These standards, which mandatorily get implemented at the planning and requirement stages, have led to security becoming intrinsic to design and manufacturing in aerospace. This is in contrast to the extrinsic security widely practiced in the IoT space—security being addressed at the final stage of the product development cycle, rather than earlier.

Standards also enabled a brand like Airbus to evolve, in competition with Boeing. Airbus began as a consortium of European firms in the 1970s, before evolving into its present form, being wholly-owned by European Aeronautic Defence and Space organisation. Currently, it manufactures across locations in Europe, China and the US.

Chandran also highlighted how standardisation has worked across the auto industry and, hence, can act as a blueprint for IoT evangelists. **EFY**